

Business Requirements Document

AI, ML and Data Jobs Dashboard

Business Objective

The business requires an interactive analytics dashboard to track and analyse job market trends across Artificial Intelligence, Machine Learning, and Data roles. The dashboard will provide comprehensive visibility into job postings, salary benchmarks, skill demand, company hiring patterns, and geographic distribution, enabling the talent acquisition and workforce planning teams to:

- Identify the most in-demand roles and skills across AI, ML, and Data domains.
- Benchmark salary ranges by experience level, work mode, and region.
- Analyse hiring patterns across companies, industries, and locations.
- Understand remote vs on-site vs hybrid work mode distribution by country.
- Support strategic workforce planning and recruitment decisions.

Scope of the Report

In Scope

- Job postings across AI, ML, and Data roles from selected companies.
- Companies covered: DataCamp, Google DeepMind, HuggingFace, NVIDIA, OpenAI.
- All experience levels: Entry, Mid, Senior.
- All employment types: Full-time, Contract, Internship.
- Work modes: Mostly Remote / Hybrid, Hybrid, On-Site, Remote.
- Geographic coverage: UK, USA, India, Canada, Germany (and other posted locations).
- Performance breakdowns by role, skill, company, location, experience level, seniority, and time period.

Dashboard Pages

The dashboard consists of four report pages:

- **1. Overview** — High-level summary of all job postings with KPI cards and top-level trend charts.
- **2. Location Insights** — Geographic breakdown of jobs, salaries, work modes, and employment types by country and region.
- **3. Company and Role Insights** — Deep-dive into hiring patterns per company, experience level, work mode, and role.
- **4. Skill and Hiring Insights** — Analysis of skill demand, industry distribution, skill usage by experience level, and salary by skill.

KPIs & Definitions

The following Key Performance Indicators are displayed as summary cards across the dashboard pages and update dynamically based on active filters:

KPI	Definition	Formula / Source	Example Use
Total Jobs	Total number of job postings in the dataset.	Count of Job_ID	Overall market volume
Average Salary	Mean salary across all filtered job postings.	Average of Salary_USD	Salary benchmarking
Median Salary	Median salary across all filtered job postings.	Median of Salary_USD	Salary distribution
Avg Salary by Exp.	Average salary filtered to a specific experience band.	Average of Salary_USD by Experience_Level	Experience benchmarking
Remote Job %	Percentage of jobs with a remote work arrangement.	$(\text{Remote Jobs} \div \text{Total Jobs}) \times 100$	Remote work trend
Total Jobs in UK	Count of job postings located in the United Kingdom.	Count of Job_ID where Country = UK	Location insight
Total Jobs in USA	Count of job postings located in the United States.	Count of Job_ID where Country = USA	Location insight
Total Jobs in India	Count of job postings located in India.	Count of Job_ID where Country = India	Location insight
Total Jobs in Canada	Count of job postings located in Canada.	Count of Job_ID where Country = Canada	Location insight
Total Jobs in Germany	Count of job postings located in Germany.	Count of Job_ID where Country = Germany	Location insight
Jobs at [Company]	Count of job postings for each specific company (DataCamp, Google DeepMind, HuggingFace, NVIDIA, OpenAI).	Count of Job_ID where Company_Name = [X]	Company hiring volume
Total Industries	Number of distinct industry categories represented.	Distinct count of Industry	Industry diversity
Total Companies	Number of distinct companies with job postings.	Distinct count of Company_Name	Competitive landscape
Total Skills	Number of distinct skills required across all postings.	Distinct count of Skill_Name	Skill breadth
Total Skills Usage Count	Sum of all skill occurrences across all job postings.	Sum of AverageSkillPerJob	Skill demand volume
Total Senior Roles	Count of job postings flagged as senior-level.	Count of Job_ID where SeniorityFlag = Senior	Seniority mix

Filters & Slicers

Filters are available across all pages and drive dynamic updates to every visual on the selected page:

Filter / Slicer	Available Options & Behaviour
Experience Level	Filters all visuals by experience band. Options: Entry, Mid, Senior. Affects KPI cards, charts, and tables across all pages.
Employment Type	Filters by contract arrangement. Options: Full-time, Contract, Internship. Available on Overview and Location Insights pages.
Work Mode	Filters by work arrangement. Options: Mostly Remote / Hybrid, Hybrid, On-Site, Remote. Used on Overview, Location Insights, and Company & Role Insights pages.
Region (Location Slicer)	Hierarchical slicer on the Location Insights page. Options: Asia, Europe, North America. Expands to country level on selection.
Company Name	Multi-select filter on Company and Role Insights. Options: DataCamp, Google DeepMind, HuggingFace, NVIDIA, OpenAI. Default: All.
Skill Name	Filter on Skill and Hiring Insights page. Allows selection of one or more skills to isolate usage patterns and associated salary data.
Posted Date / Month	Time filter linked to Dim_Date. Allows filtering visuals by posting month or date range.

Charts Requirements

Page 1: Overview

The Overview page gives a high-level snapshot of the entire job market dataset with five top KPI cards and three charts.

- **Total Jobs by Month**

An area chart showing the count of job postings over time.

- X-axis: Month (from Dim_Date[MonthName]).

- Y-axis: Total Jobs (Count of Job_ID).

- Purpose: Identify hiring peaks and seasonal trends in job postings.

- **Total Employees by Work Mode**

A clustered bar chart showing headcount split by work arrangement.

- X-axis: Work Mode categories (Mostly Remote / Hybrid, Hybrid, On-Site, Remote).

- Y-axis: Employee Count.

- Purpose: Understand the distribution of work flexibility across the dataset.

- **Job Posted by Experience Level**

A donut chart showing the proportion of jobs per experience band.

- Segments: Mid, Entry, Senior.

- Labels show count and percentage per segment.

- Purpose: Understand the seniority mix of available roles.

- **Experience Level Distribution by Employment Type**

A horizontal stacked bar chart comparing employment type across experience levels.

- Y-axis: Experience Level (Mid, Entry, Senior).
- X-axis: Number of Jobs.
- Stacks: Employment Type — Full-time, Contract, Internship.
- Purpose: Reveal how employment types vary by experience band.

Page 2: Location Insights

The Location Insights page provides a geographic breakdown of job postings, salary benchmarks, and work mode patterns across countries and regions.

- **Total Jobs with Avg. Salary in Each Country**

A treemap visualising job volume and average salary by country and city.

- Size of each tile: Total job count for that country / city.
- Hierarchy: Country → City.
- Purpose: Compare hiring volumes and salary levels across geographies at a glance.

- **Employment Type Distribution across Countries**

A horizontal stacked bar chart showing the split of employment types per country.

- Y-axis: Country (UK, USA, India, Canada, Germany).
- X-axis: Number of Jobs.
- Stacks: Employment Type — Contract, Full-time, Internship.
- Purpose: Identify which countries prefer certain employment arrangements.

- **Employees Work Mode by Country**

A map visual with pie chart overlays showing work mode distribution per country.

- Bubble / pie chart location: Country centroid on world map.
- Pie segments: Work Mode breakdown (Hybrid, Mostly Remote, On-Site, Remote).
- Purpose: Provide an at-a-glance geographic view of remote and on-site trends.

- **Region wise Avg. Salary by Experience**

A matrix table showing average salary broken down by region and experience level.

- Rows: Region (North America, Europe, Asia).
- Columns: Experience Level (Entry, Mid, Senior).
- Values: Average Salary_USD.
- Purpose: Benchmark compensation expectations by region and seniority.

Page 3: Company and Role Insights

This page provides a detailed breakdown of hiring patterns, role distribution, and salary trends across the five featured companies.

- **Jobs Per Company by Experience Level**

A 100% stacked bar chart showing the experience level composition per company.

- X-axis: Company Name (DataCamp, Google DeepMind, HuggingFace, NVIDIA, OpenAI).
- Y-axis: % of Jobs Per Company.
- Stacks: Experience Level — Entry, Mid, Senior.
- Purpose: Compare the seniority mix across companies.

- **Jobs by Experience Level and Work Mode**

A small-multiples scatter / dot chart broken into four quadrants by experience level.

- Panels: Blank, Entry, Mid, Senior (one panel per experience level).
- X-axis: Job Title.
- Y-axis / markers: Work Mode (Hybrid, Mostly Remote / Hybrid, On-Site, Remote).
- Purpose: Understand how role titles and work modes intersect across experience levels.

- **Companies Rating Comparison**

A horizontal bar chart comparing the average company rating for each employer.

- Y-axis: Company Name.
- X-axis: Average Company Rating (Comp. Rating).
- Each bar is colour-coded per company.
- Purpose: Benchmark employer attractiveness across the featured companies.

- **Job Remote Ratio with Skill Usage by Avg. Salary**

A bubble / scatter chart plotting remote ratio against average salary, sized by skill usage.

- X-axis: Sum of Remote_Ratio (0–100).
- Y-axis: Average Salary_USD.
- Each bubble represents a Job Title, labelled on the chart.
- Purpose: Identify high-paying, remote-friendly roles with high skill requirements.

Page 4: Skill and Hiring Insights

This page analyses the demand for technical skills, their distribution across industries and experience levels, and their relationship with compensation.

- **Skill Usage by Experience Level**

A 100% stacked bar chart showing how each skill is distributed across experience bands.

- X-axis: Skill Name (Python, Machine Learning, Cloud Computing, etc.).
- Y-axis: Skill Usage % per experience level.
- Stacks: Experience Level — Entry, Mid, Senior.
- Purpose: Identify skills more prevalent at specific seniority levels.

- **Industry wise Skill Usage**

A pie chart showing the share of total skill usage per industry.

- Segments: AI/ML, AI Research, EdTech, Hardware, NLP.
- Labels show count and percentage for each industry.
- Purpose: Understand which industries drive the most skill demand.

- **Avg. Salary for Jobs by Seniority Flag**

An area / waterfall chart showing average salary by job title, segmented by seniority.

- X-axis: Job Title.
- Y-axis: Average Salary_USD.
- Segments: Non Senior vs Senior (SeniorityFlag).
- Purpose: Highlight salary premiums for senior-flagged roles across job titles.

• Skills and Average Salary by Work Mode

A treemap visualising skills grouped and sized by work mode and average salary.

■ Hierarchy: Work Mode → Skill Name.

■ Tile size: Frequency / usage count of the skill within that work mode.

■ Purpose: Reveal which skills are most associated with specific work arrangements and their corresponding salary levels.

• Jobs Wise Skill Usage

A horizontal bar chart showing total skill usage count per job title.

■ Y-axis: Job Title (ML Engineer, AI Product Manager, Computer Vision Engineer, etc.).

■ X-axis: Skill Usage count.

■ Bars are colour-coded by job category.

■ Purpose: Identify which roles demand the broadest or highest skill requirements.

Data Model Summary

The dashboard is powered by a star schema consisting of two fact tables and four dimension tables. All relationships are defined in Power BI's model view:

Table	Type	Description	Key Fields
Fact_Jobs	Fact	Core fact table. One row per job posting.	Job_ID, Company_ID, Location_ID, Job_Title, Employment_Type, Experience_Level, Salary_USD, Salary_Group, Remote_Ratio, SeniorityFlag, PostingMonth, Posted_Year-Month, Posted_Date
Fact_Job_Skills	Fact	Skill-level fact table. One row per job-skill combination.	Job_ID, Skill_ID, AverageSkillPerJob
Dim_Companies	Dimension	Company master data.	Company_ID, Company_Name, Company_Size, Industry, Rating
Dim_Locations	Dimension	Location master data.	Location_ID, City, Country, Region
Dim_Skills	Dimension	Skill master data.	Skill_ID, Skill_Name
Dim_Date	Dimension	Date / calendar table for time intelligence.	Date, DayName, DayNum, MonthName, MonthNum

Relationships are defined as follows: Fact_Jobs links to Dim_Companies via Company_ID (many-to-one), to Dim_Locations via Location_ID (many-to-one), and to Dim_Date via Posted_Date (many-to-one). Fact_Job_Skills links to Fact_Jobs via Job_ID (many-to-one) and to Dim_Skills via Skill_ID (many-to-one).

Key Calculated Measures

The following DAX measures are required to power the KPI cards and chart calculations:

Measure Name	DAX Formula / Logic
Total Jobs	COUNT(Fact_Jobs[Job_ID])

Measure Name	DAX Formula / Logic
Average Salary	AVERAGE(Fact_Jobs[Salary_USD])
Median Salary	MEDIAN(Fact_Jobs[Salary_USD])
Remote Job %	DIVIDE(CALCULATE([Total Jobs], Fact_Jobs[Remote_Ratio] = 100), [Total Jobs]) * 100
Avg Salary by Exp	CALCULATE(AVERAGE(Fact_Jobs[Salary_USD]), ALLEXCEPT(Fact_Jobs, Fact_Jobs[Experience_Level]))
Total Senior Roles	CALCULATE([Total Jobs], Fact_Jobs[SeniorityFlag] = "Senior")
Total Skills Usage	SUM(Fact_Job_Skills[AverageSkillPerJob])
Jobs in [Country]	CALCULATE([Total Jobs], Dim_Locations[Country] = "[Country]")
Jobs at [Company]	CALCULATE([Total Jobs], Dim_Companies[Company_Name] = "[Company]")
Avg Salary by Region	CALCULATE(AVERAGE(Fact_Jobs[Salary_USD]), ALLEXCEPT(Dim_Locations, Dim_Locations[Region]))